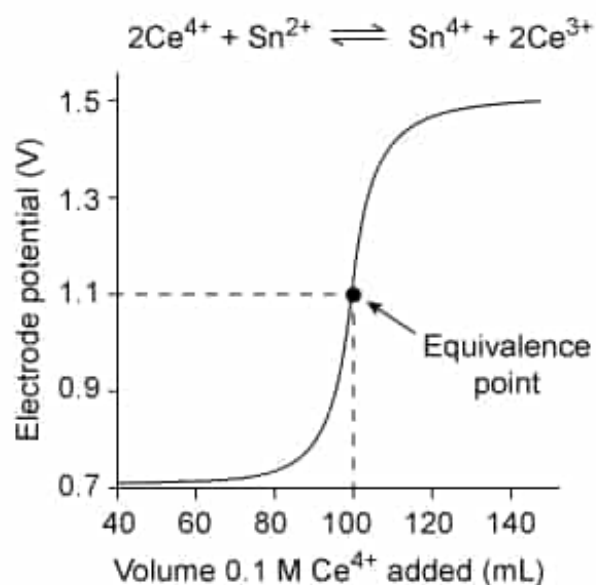


The graph below shows the redox titration curve of a 0.05 M Sn^{2+} solution titrated with 0.1 M Ce^{4+} .



The equivalence point indicates the volume of 0.1 M Ce^{4+} required to fully oxidize all of the Sn^{2+} in solution. If more than 100 mL of 0.1 M Ce^{4+} is required to titrate an equal volume of another 0.05 M Sn^{2+} solution, which of the following could be used to explain the inaccurate equivalence point?

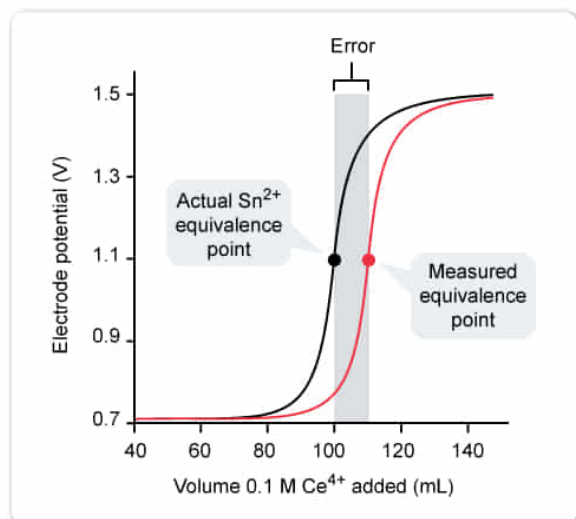
- ☐ A. The volume of the Sn^{2+} solution has increased. [7%]
- ☐ B. The concentration of Sn^{2+} in solution has increased. [24%]
- ☐ C. An impurity that can oxidize Ce^{3+} back to Ce^{4+} is present in the Sn^{2+} solution. [20%]
- ☒ D. A metal salt impurity that can also be oxidized by Ce^{4+} is present in the Sn^{2+} solution. [46%]

Correct

47% answered correctly

Explanation:

Decreased equivalence point accuracy
due to an oxidized impurity

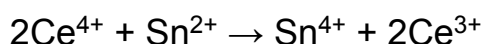


- Oxidation of the impurity consumes additional Ce^{4+}
- Sn^{2+} and the impurity are indistinguishable

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During a **redox titration**, a solution with a known concentration of an oxidizing agent (such as Ce^{4+}) is slowly added to a solution with an unknown concentration of a species to be measured (**analyte**). As the oxidizing agent is added, the analyte is oxidized and the electrode potential changes. The voltage sharply rises when passing through the **equivalence point**, which occurs when all of the analyte in the solution has been oxidized.

In the given titration of the Sn^{2+} analyte, 100 mL of 0.1 M Ce^{4+} are required to reach the equivalence point. At that point, all of the Sn^{2+} has been oxidized to Sn^{4+} according to the following reaction:



If only Sn^{2+} is being oxidized in the titration, then the consumption of 100 mL of 0.1 M Ce^{4+} correlates exclusively to the concentration of Sn^{2+} . However, if a metal salt impurity that could also be oxidized by Ce^{4+} was in the solution, then the volume of 0.1 M Ce^{4+} consumed would be due to both the oxidation of Sn^{2+} and the oxidation of the metal impurity.

The oxidation of a metal impurity would consume an additional amount of Ce^{4+} solution beyond the 100 mL required to fully oxidize the Sn^{2+} . As such, the measured equivalence point would occur at a greater volume than the 100 mL of 0.1 M Ce^{4+} that is attributed solely to the oxidation of Sn^{2+} . Because the oxidation of Sn^{2+} and the impurity **cannot be distinguished** in the titration, this introduces error

and **decreases the accuracy** of the measured value of the equivalence point.

(Choices A and B) The question states that the volume and Sn^{2+} concentration is the same in both of the titrated solutions.

(Choice C) An impurity that could oxidize Ce^{3+} back to Ce^{4+} would regenerate some of the added titrant, and this would mean that *less than 100 mL* (not more) would be needed to reach the equivalence point.

Educational objective:

Impurities that undergo the same oxidation-reduction behavior as the analyte cannot be distinguished from the analyte and decrease the accuracy of the measured equivalence point in a redox titration.

Time spent:0 sec

Subject:

General Chemistry

QID:402132

System:

Solutions and Electrochemistry